

Disproof of Conjecture 00000001685: no even subdivision of $K_{3,3}$ is a Pfaffian graph

jilint777

2026-10-04

Abstract

Conjecture 00000001685 says that the minimal families of nonplanar Pfaffian graphs are the Möbius ladders M_{2k} and the even subdivisions of $K_{3,3}$. In particular it says that the even subdivisions of $K_{3,3}$ are Pfaffian graphs. This is false. We prove that *no* even subdivision of $K_{3,3}$ has a Pfaffian orientation, including $K_{3,3}$ itself, which is also the Möbius ladder on 6 vertices. The proof is a parity argument: in every orientation an odd number of the six perfect-matching terms of the Pfaffian are negative, so the terms never all have the same sign. The obstruction, and its instances for $K_{3,3}$, for a proper even subdivision of $K_{3,3}$, and for the 6-vertex Möbius ladder, are formalized in Lean 4 (core library only). As a secondary result we give a nonplanar Pfaffian graph of crossing number 1 that lies in neither family.

1 The conjecture and the clause we refute

Definition: A Pfaffian orientable graph is one whose perfect matchings can be counted by signed cancellation. *Conjecture:* The minimal families of nonplanar Pfaffian graphs are the Möbius ladders M_{2k} and the even subdivisions of $K_{3,3}$; these two families exhaust the nonplanar Pfaffian class of crossing number ≤ 2 .

The Chinese version says the same. The first sentence names two families *of nonplanar Pfaffian graphs*, and the second calls them part of “the nonplanar Pfaffian class”. So the conjecture asserts, at the least:

Clause (E). Every even subdivision of $K_{3,3}$ is a (nonplanar) Pfaffian graph.

The conjecture is a conjunction that includes (E), so refuting (E) refutes the conjecture. Whatever “minimal” means, a family of nonplanar Pfaffian graphs has to consist of Pfaffian graphs, and the falsity of (E) does not depend on that word.

Readings covered.

- *Even subdivision.* An even subdivision of H replaces every edge by a path with an even number $(0, 2, 4, \dots)$ of internal vertices. We prove (Theorem 7) that *every* even subdivision of $K_{3,3}$ is non-Pfaffian. So it makes no difference whether $K_{3,3}$ itself counts as an even subdivision, whether at least one edge must be subdivided, or whether every edge must be subdivided.

- *Pfaffian*. We use the conjecture's own definition (the signed sum counts the perfect matchings, $|\text{Pf}(A_D)| = \#\text{PM}$) and Kasteleyn's standard definition (all perfect matchings have the same sign). They are equivalent (Lemma 2), and both are refuted.
- M_{2k} . If M_{2k} is the Möbius ladder on $2k$ vertices (k rungs), then $M_6 \cong K_{3,3}$ is not Pfaffian, so the Möbius part fails as well. More generally, the computations below show that the ladders with 3, 5 and 7 rungs are non-Pfaffian. If instead M_{2k} has $2k$ rungs, the ladders with 2, 4 and 6 rungs are Pfaffian, and we make no claim against the Möbius part. In both cases the refutation of (E) does not depend on this choice.

2 Definitions

Let G be a loopless graph on the vertex set $\{1, \dots, n\}$ (we use $\{0, \dots, n-1\}$ in the code). A *perfect matching* is a set of edges covering every vertex exactly once. An *orientation* D directs every edge, and its skew adjacency matrix is $A_D = (a_{ij})$, where $a_{ij} = 1$ if $i \rightarrow j$, $a_{ij} = -1$ if $j \rightarrow i$, and $a_{ij} = 0$ otherwise. Let $M = \{t_1 h_1, \dots, t_m h_m\}$ be a perfect matching ($n = 2m$), with each edge written tail-first ($t_k \rightarrow h_k$). Its *term* is

$$\text{sgn}_D(M) = \text{sgn} \begin{pmatrix} 1 & 2 & \cdots & 2m-1 & 2m \\ t_1 & h_1 & \cdots & t_m & h_m \end{pmatrix} = (-1)^{\text{inv}(t_1 h_1 \cdots t_m h_m)},$$

where inv counts inversions. This does not depend on the order in which the edges are listed: moving a block of two entries past another entry is an even permutation. The Pfaffian is

$$\text{Pf}(A_D) = \sum_{\pi} \text{sgn}(\pi) \prod_k a_{\pi(2k-1)\pi(2k)} = \sum_M \text{sgn}_D(M).$$

Here π runs over the permutations with $\pi(1) < \pi(3) < \dots$ and $\pi(2k-1) < \pi(2k)$. A term is non-zero exactly when the pairs form a perfect matching M . Writing each pair tail-first changes the sign of the permutation and of the factor a together, so the term equals $\text{sgn}_D(M)$.

Definition 1. D is a *Pfaffian orientation* if $|\text{Pf}(A_D)|$ equals the number $\Phi(G)$ of perfect matchings. G is *Pfaffian* if it has a Pfaffian orientation.

Lemma 2. $|\text{Pf}(A_D)| = \Phi(G)$ if and only if all perfect matchings have the same sign $\text{sgn}_D(M)$.

Proof. Suppose p of the Φ terms are $+1$ and q are -1 . Then $|\text{Pf}| = |p - q|$, and this equals $p + q$ if and only if $pq = 0$. $\square$

3 The parity obstruction

Lemma 3. Reversing one edge e of D changes $\text{sgn}_D(M)$ for exactly the perfect matchings $M \ni e$.

Proof. If $e \notin M$, the word $t_1 h_1 \cdots t_m h_m$ is unchanged. If $e = t_k h_k \in M$, the two adjacent entries t_k, h_k are swapped. They are distinct because G is loopless, so the swap is a transposition and the inversion number changes by exactly 1. $\square$

Theorem 4 (Parity obstruction). *Let G be loopless, and suppose that*

- (a) $\Phi(G)$ is even;
- (b) every edge lies in an even number of perfect matchings;

(c) for some orientation D_0 , the number of perfect matchings M with $\text{sgn}_{D_0}(M) = -1$ is odd.

Then G is not Pfaffian.

Proof. Let $\nu(D)$ be the number of negative terms. Any orientation D is obtained from D_0 by reversing edges one at a time. By Lemma 3, reversing e changes ν by an amount congruent modulo 2 to the number of perfect matchings containing e , which is even by (b). Hence $\nu(D) \equiv \nu(D_0) \equiv 1 \pmod{2}$ for every D . If D were Pfaffian, Lemma 2 would give $\nu(D) \in \{0, \Phi\}$, which is even by (a). This is a contradiction. $\square$

Equivalently, by (b) the product $\prod_M \text{sgn}_D(M) = (-1)^{\nu(D)}$ does not depend on D . It is -1 , while a Pfaffian orientation would make it $(\pm 1)^\Phi = 1$. This is the classical reason why $K_{3,3}$ is not Pfaffian (Little [1]; see also [2, 3]).

4 $K_{3,3}$

Let the vertices be $0, 1, 2$ and $3, 4, 5$, and let D_0 orient every edge $a \rightarrow b$ with $a \in \{0, 1, 2\}$. The perfect matchings are $M_\sigma = \{0\sigma(0), 1\sigma(1), 2\sigma(2)\}$ for the six bijections $\sigma : \{0, 1, 2\} \rightarrow \{3, 4, 5\}$. In the word $0\sigma(0)1\sigma(1)2\sigma(2)$ the entries $\sigma(i) \geq 3$ form $2 + 1 = 3$ inversions with the later entries $1, 2$, plus the inversions of σ itself, so $\text{sgn}_{D_0}(M_\sigma) = -\text{sgn}(\sigma)$:

matching	mask (Lean)	$\text{sgn}(\sigma)$	sgn_{D_0}
$\{05, 14, 23\}$	84	-1	$+1$
$\{04, 15, 23\}$	98	$+1$	-1
$\{05, 13, 24\}$	140	$+1$	-1
$\{03, 15, 24\}$	161	-1	$+1$
$\{04, 13, 25\}$	266	-1	$+1$
$\{03, 14, 25\}$	273	$+1$	-1

(Bit i of a mask selects edge number i in the order $03, 04, 05, 13, 14, 15, 23, 24, 25$.) Thus $\Phi = 6$; each edge ab lies in exactly 2 matchings (the 2 bijections with $\sigma(a) = b$); and $\nu(D_0) = 3$ is odd. By Theorem 4:

Proposition 5. $K_{3,3}$ is not Pfaffian. Concretely, $\text{Pf}(A_{D_0}) = 0$, and every one of the $2^9 = 512$ orientations has $|\text{Pf}(A_D)| \leq 4 < 6$.

(Indeed $\nu(D)$ is odd, so $|\text{Pf}| = |6 - 2\nu| \in \{4, 0\}$.)

5 All even subdivisions of $K_{3,3}$

Let G have vertices $\{0, \dots, n-1\}$ and an edge uv . Let G' be the graph obtained by replacing uv with the path $uabv$, where $a = n$ and $b = n+1$ are new vertices. Every even subdivision of $K_{3,3}$ arises from $K_{3,3}$ by finitely many such steps, up to relabelling of the vertices.

Lemma 6. If G satisfies (a), (b) and (c) of Theorem 4, so does G' , and $\Phi(G') = \Phi(G)$.

Proof. In G' the vertex a is matched either to b or to u ; in the latter case b is matched to v . This gives a bijection $M \mapsto M'$ from the perfect matchings of G :

$$M' = M \cup \{ab\} \text{ if } uv \notin M, \quad M' = (M \setminus \{uv\}) \cup \{ua, bv\} \text{ if } uv \in M.$$

So $\Phi(G') = \Phi(G)$ and (a) is inherited. Let c be the number of perfect matchings of G through uv . In G' the edges ua and bv lie in c perfect matchings and ab lies in $\Phi - c$. Both numbers are even, and the other edges keep their counts, so (b) is inherited. For (c): since G satisfies (b), the proof of Theorem 4 shows that $\nu(D)$ is odd for *every* orientation D of G , so we may take D_0 with $u \rightarrow v$. Orient $u \rightarrow a \rightarrow b \rightarrow v$ and keep every other edge. If $uv \notin M$, the word of M' is the word of M followed by ab . These are the two largest labels, in increasing order, so the sign is unchanged. If $uv \in M$, list uv last, so that the word of M is Wuv . The word of M' is $Wuabv$. It arises from $Wuvab$ (which has the sign of Wuv) by moving v past a and b , an even permutation. Hence $\text{sgn}_{D'_0}(M') = \text{sgn}_{D_0}(M)$ for all M , and ν is preserved. $\square$

Relabelling the vertices by a permutation τ multiplies every term by $\text{sgn}(\tau)$. So conditions (a), (b) and (c) are invariant under relabelling: when Φ is even, ν either stays the same or becomes $\Phi - \nu$. Being Pfaffian is also invariant. Induction with Lemma 6, starting from Proposition 5, gives:

Theorem 7. *Every even subdivision of $K_{3,3}$ (including $K_{3,3}$ itself) satisfies the hypotheses of Theorem 4 with $\Phi = 6$. Hence no even subdivision of $K_{3,3}$ is Pfaffian, and Clause (E) of Conjecture 00000001685 is false.*

Corollary 8. *Conjecture 00000001685 is false.*

The smallest proper even subdivision is S : $K_{3,3}$ with edge 03 replaced by the path 0673 (8 vertices, 11 edges). It is non-Pfaffian with $\Phi(S) = 6$. The masks of its perfect matchings, over the edge order 04, 05, 13, 14, 15, 23, 24, 25, 06, 67, 73, are 554, 561, 582, 645, 1360, 1416.

6 Möbius ladders

The Möbius ladder with r rungs has vertices $0, \dots, 2r - 1$, the cycle edges $i(i + 1 \bmod 2r)$ and the rungs $i(i + r)$. For $r = 3$ it is $K_{3,3}$, with colour classes the even and the odd vertices. Its 6 perfect matchings satisfy the obstruction directly; this is proved in Lean as `M6_not_pfaffian`. An exhaustive computation in `verify.py`, using the method of the next section, gives the following:

rungs r	2	3	4	5	6	7
vertices	4	6	8	10	12	14
Pfaffian?	yes	no	yes	no	yes	no

The ladder with 5 rungs has 13 perfect matchings, and none of its 2^{15} orientations is Pfaffian. In the vertex-count convention, then, the Möbius part of the conjecture also fails (for example at M_6 and M_{10}). In the rung-count convention we make no claim about it.

7 Secondary: the second sentence

Read the second sentence as “every nonplanar Pfaffian graph of crossing number ≤ 2 is a Möbius ladder or an even subdivision of $K_{3,3}$ ”. Under this reading it is also false. Let W be the Wagner graph V_8 (the Möbius ladder with 4 rungs) with the edge 01 replaced by a path 0891. Then:

- W is Pfaffian, with $\Phi(W) = 7$. The proof of Lemma 6 shows that the compatible orientation of a Pfaffian orientation of V_8 is Pfaffian. The Lean theorem `M8_pfaffian` exhibits a Pfaffian orientation of V_8 , and `verify.py` checks W directly.

- W is nonplanar. $V_8 - \{04\}$ is a subdivision of $K_{3,3}$ with classes $\{1, 3, 6\}$ and $\{2, 5, 7\}$: the paths are 12, 15, 107, 32, 37, 345, 65, 67, 62.
- $\text{cr}(W) \leq 1$. Draw the ladder with rails 0123 (top) and 4567 (bottom), and with vertical rungs. Draw the edge 34 around the top and down the left side, and draw 70 around the bottom and up the left side. The two edges cross exactly once. Subdividing an edge does not change the drawing.
- W is not a Möbius ladder, since it has vertices of degree 2. It is not an even subdivision of $K_{3,3}$, since $|E| - |V| = 14 - 10 = 4$, whereas every subdivision of $K_{3,3}$ has $|E| - |V| = 3$.

This part is not formalized in Lean, apart from `M8_pfafrican`. The main disproof does not depend on it.

8 Independent computational check (verify.py)

The script uses only the Python standard library and a different method from the Lean proof. Since $\det A_D = \text{Pf}(A_D)^2$ and $|\text{Pf}| \leq \Phi$, G is Pfaffian if and only if some orientation has $\det A_D = \Phi^2$. The script computes exact integer determinants (Bareiss elimination) and enumerates perfect matchings recursively. It checks:

- $K_{3,3}$ (all 512 orientations), S (all 2048), and $K_{3,3}$ with two edges subdivided twice (all 8192): none is Pfaffian, and $\Phi = 6$ in each case.
- $K_{3,3}$ with every edge subdivided 2 times, and with every edge subdivided 4 times. Switching at a vertex (reversing all its edges) multiplies the Pfaffian by -1 and leaves $\det$ unchanged. So it suffices to check one orientation per switching class: fix the edges of a spanning tree, which leaves $2^{|E|-|V|+1} = 16$ orientations. None is Pfaffian.
- The hypotheses (a), (b), (c) of Theorem 4 for all 2^9 even subdivisions with 0 or 2 internal vertices per edge, and for three more with up to 6.
- Möbius ladders with 2–7 rungs (the table above), with 5 rungs also exhaustively over 2^{15} orientations; and the graph W of the previous section.

9 Lean formalization

`lean4/` is a self-contained Lean 4.19.0 project (core library only, no Mathlib). Everything is defined from scratch:

- **Graph** is a vertex count and an edge list. `sel es M` is the edge subset selected by the bit mask M . `isPM` says every vertex $v < n$ lies on exactly one selected edge. `PMs G` lists all perfect matchings among the $2^{|E|}$ masks, and `numPM G` is their number.
- An orientation is a `List Bool`, one entry per edge (`true` keeps (a, b) as $a \rightarrow b$). `par` is the inversion parity of the word $t_1 h_1 t_2 h_2 \dots$, and `term` is ± 1 accordingly. `Pf G os =  $\sum_M \text{term}$` .
- `IsPfaffian G`: some orientation of length $|E|$ has $|\text{Pf}| = \text{numPM}$ (the conjecture’s definition). `IsPfaffian'`: all terms are equal (Kasteleyn’s definition). `same_sign_pfafrican` shows that the second implies the first.

- `not_pfaffian_of_invariant` is Theorem 4 for arbitrary graphs. It is proved through `word_perm`, `par_flips` (Lemma 3 in cumulative form: the parity changes by the number of reversed matching edges), `sum_map_flips_even` and `sum_terms`.
- Computations (kernel `decide`): `PMs_K33`, `PMs_S`, `PMs_M6`, and the hypotheses (a)–(c). The results are `K33_not_pfaffian`, `S_not_pfaffian` and `M6_not_pfaffian`.
- `EvenSubdivisionK33` is the inductive family generated from $K_{3,3}$ by `subdivide2`, which replaces an edge uv by $un(n+1)v$. `S_even_subdivision` shows S belongs to it and $S \neq K_{3,3}$.
- Main theorems: `conjecture_00000001685_false` (Q): $\neg \forall G, \text{EvenSubdivisionK33 } G \rightarrow Q \wedge \text{IsPfaffian } G$, for an arbitrary extra property Q such as “nonplanar” or “minimal”; the same with `IsPfaffian'`; `_false_proper` (members other than $K_{3,3}$); `_conjunction_false` ($\neg((E) \wedge R)$ for any R); and `mobius_clause_false_vertex_convention`.
- Non-vacuity: `K4_pfaffian` (K_4 , the ladder with 2 rungs) and `M8_pfaffian` (the nonplanar Wagner graph) have explicit Pfaffian orientations, so `IsPfaffian` is not empty. As a cross-check that `Pf` is the matrix `Pfaffian`, `pfMat_K33` and `pfMat_M8` compare it with the recursive row expansion $\text{Pf}(A) = \sum_j (-1)^j a_{1j} \text{Pf}(A_{1\hat{j}})$ of the skew matrix.

Not in Lean. Three things are proved here but not in Lean: the induction of Lemma 6 for arbitrary even subdivisions (Lean covers $K_{3,3}$ and S , which already refute every reading of (E)); the invariance under relabelling; and Section 7 apart from `M8_pfaffian`. There is no `sorry`, no `native_decide`, and no added axiom. `#print axioms` shows at most `propext` and `Quot.sound`.

10 Reproduction

```
cd lean4 && lake build      # Lean 4.19.0, core only (about 1 minute)
cd .. && python3 verify.py # Python 3 standard library
pdflatex report.tex && pdflatex report.tex
```

References

- [1] C. H. C. Little, A characterization of convertible $(0, 1)$ -matrices, *J. Combin. Theory Ser. B* 18 (1975), 187–208.
- [2] L. Lovász and M. D. Plummer, *Matching Theory*, North-Holland, 1986.
- [3] N. Robertson, P. D. Seymour and R. Thomas, Permanents, Pfaffian orientations, and even directed circuits, *Ann. of Math.* 150 (1999), 929–975.